Herbalism

Game Design Documentation

Potionsmith

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One Page

Elevator Pitch

A single-player crafting and management simulation strategy game where you play as an alchemist, brewing potions, refining ingredients, and fulfilling customer requests to build your reputation and grow your potion shop.

Category

Single Player / Crafting / Shop Management / Time Management / Simulation

Overview

Herbalism is a game built on the back of tying the game *Potion Craft* together with a more in-depth order fulfillment-based game, such as *Papa's Pizzeria* or *Papers Please*. The game places particular emphasis on tactile gameplay and explicit step following while using documents as its primary source of instruction. The player is set in the world of Ethshar, and this world is specifically emphasized to further develop a sense of immersion.

Play Value

The main play value of the game comes from its hands-on, exploratory crafting experience, which rewards curiosity and experimentation. By selecting, refining, and combining ingredients, players engage in tactile decision-making that directly influences the final potion. Each ingredient has predefined refinement paths and behaviors outlined in the recipe book, guiding the player through the required steps for each potion. This process fosters a strong sense of discovery and mastery as players learn how their actions affect potion quality. The gameplay loop of choosing ingredients, refining them with different tools, mixing them in the cauldron, and racing against the timer creates an immersive and satisfying rhythm that keeps players engaged throughout each day in the shop.

Vision Statement

Herbalism is a game about brewing potions for clients and placing the player into the shoes of an alchemist in the world of Ethshar.

Narrative

Theme

The theme is focused around the cozy vibes of a home-owned potion shop. The simple but homely feel from the assets and environment all contribute to reinforcing this overall mood. Running a shop can be hectic, but at the end of the day, brewing potions for customers is supposed to feel somewhat relaxing and fulfilling.

Characters

The game does not have much of a cast. Outside of the Wizard who passes the shop off to you, all other npcs are merely there for informational dialogue about what potion they may want. The Wizard is not too fleshed out either, but wizards in the world of Ethshar can tend to be a little… scattered. So his introduction at the start of the game is merely him getting bored with running the shop so he decides to leave it in the capable hands of his unprepared apprentice.

Story

You are a Wizard's apprentice in the world of Ethshar and the Wizard has gotten bored of running his old potion brewing shop. His solution? Pass it off to you, with little to no instructions, and merely a book of potion recipes to aid you. You must keep the potion shop running until he returns from goodness knows, whatever he's doing.

World

Your shop exists in the Ethshar of the Spices. The world is not incredibly relevant as it's outside the doors of your shop. Aside from the occasional relevant potion request, this does not serve much of an impact aside from mechanical world building (port trading town is great for acquiring potion ingredients).

Onboarding

The onboarding is completely diegetic through a tutorial level with the former owner of the shop handing it off and giving us a step-by-step tutorial on how to brew a potion. This is a simple luck potion, with the wizard that formerly ran the shop ordering it, and instructing us on how to use the book of instructions, refine ingredients, and finally complete, brew and sell a potion.

Game Balancing

The balancing for the game as its single player was never incredibly intense, however it was down to keeping our players engaged. We wanted our players to think about or do more than one thing at once, so we decided to add in as the last two refinement options, two implements that don't need to be interacted with for the whole of their refining process, such as the infuser and drying rack. This helped to up the players cognitive load a bit to work closer in line with usual shop development games.

Inspirational Media

Potion Craft

Genre:

Simulation / Crafting

Gameplay:

Players take on the role of an alchemist who experiments with various ingredients to brew potions. They manually grind herbs, mix liquids, control heating, and navigate a potion map to discover new recipes. The player interacts with customers, fulfilling potion requests to earn gold and reputation.

Resources:

- Ingredients (herbs, mushrooms, minerals)
- Water, oil, and other bases
- Gold (for trading and upgrades)
- Reputation and popularity (affecting customer trust and pricing)

Obstacles:

- Limited ingredient supply and gold
- Failed potion experiments waste materials
- Demanding or impatient customers
- Balancing moral choices (helping good or bad clients)

Play Value:

Potion Craft offers a relaxing, tactile crafting experience with a strong sense of discovery and experimentation. Its charm lies in the freedom to create, trade, and experiment at your own pace, rewarding curiosity and creativity.

Inspirations for Our Game:

- Potion crafting

Papa's Pizzeria

Genre:

Time Management / Simulation

Gameplay:

Players run a pizzeria where they take customer orders, prepare dough, add toppings, bake pizzas, and deliver them efficiently.

Resources:

- Ingredients (dough, sauce, cheese, toppings)
- Kitchen tools and ovens
- Money (to upgrade the pizzeria)
- Customer orders (as "requests")

Obstacles:

- Time pressure to fulfill multiple orders simultaneously
- Increasing complexity of customer requests

- Limited workspace and tools
- Balancing quality vs. speed

Play Value:

Fast-paced, satisfying, and addictive due to the combo of time management, multitasking, and visual feedback. Offers a sense of progression as players unlock upgrades, new ingredients, and more challenging customers.

Inspirations for Our Game:

- Order taking
- Customer orders
- Gameplay loop

Papers Please

Genre:

Puzzle / Simulation / Dystopian / Strategy

Gameplay:

Players take on the role of a border control officer in a fictional totalitarian country. They inspect travelers' documents, check for discrepancies, and decide whether to allow or deny entry.

Resources:

- Passports, ID cards, and other documents
- Rules and inspection tools (stamps, interrogation prompts, fingerprint checks)
- Money for the family (earned by correctly processing entrants)
- Information sheets (new regulations, directives, and updates)

Obstacles:

- Increasingly complex documents and rules

- Moral dilemmas (helping desperate people vs. following orders)
- Limited time to process travelers
- Penalties for mistakes, which can affect your salary and family welfare

Play Value:

Highly engaging for players who enjoy strategic thinking, pattern recognition, and moral decision-making. Creates tension through time pressure and consequences for choices, making each day feel meaningful.

Inspirations for Our Game:

- Scene layout design

Plants vs. Zombies

Genre:

Tower Defense / Strategy / Casual

Gameplay:

Players strategically place plants with different abilities along lanes to stop waves of incoming zombies from reaching their house.

Resources:

- Sun points (currency for planting)
- Various plant types (offensive, defensive, or utility)
- Garden space (limited slots per lane)

Obstacles:

- Waves of increasingly tough zombies with unique abilities
- Limited sun resources to deploy plants
- Space constraints in the garden lanes

- Timing and placement decisions under pressure

Play Value:

Engaging for both casual and strategic players. Offers a satisfying mix of planning, quick thinking, and progression with new plants and zombie types. Its charm comes from humorous visuals and escalating challenges.

Inspirations for Our Game:

- Seed selection phase before the combat

Good Pizza, Great Pizza

Genre:

Time Management / Simulation / Casual

Gameplay:

Players run a pizza shop, taking customer orders and assembling pizzas according to specific requests. Gameplay involves preparing dough, adding toppings, baking, and delivering pizzas while managing customer satisfaction.

Resources:

- Ingredients (dough, sauce, cheese, toppings)
- Kitchen tools and ovens
- Money (to buy ingredients and upgrade the shop)
- Customer orders (as “requests”)

Obstacles:

- Increasingly complex customer orders
- Time constraints to complete orders
- Balancing quality vs. speed for customer satisfaction
- Managing finances and limited resources

Play Value:

Fun, fast-paced, and addictive for players who enjoy multitasking and time management. Offers a satisfying loop of fulfilling orders, earning money, and upgrading the shop, combined with colorful visuals and humor.

Inspirations for Our Game:

- Customer UI
- Time constraint
- End of the day report

Market Analysis

Audience

People who want to feel immersed in the role of an alchemist, as well as people who like time crunch, or management games such as the Papa's series.

Platform

PC. The demand for mouse controls makes this platform the easiest to develop this style of game for.

Competition Research

The biggest competition would be *Potion Craft* itself. Though potion games do exist, they have a nasty habit of feeling intangible, so one goal would be to see or get to interact with the effects of the potions themselves.

Project Plan

Scope

For the final product, the project aims to consist of a tutorial and at least one level. This includes full mechanical interactions with each machine as well as utility for all included ingredients. For the final version, we need at least 5 refinements and 4 ingredients, and at least 3 potion effects with an amplified version for each. At least two customers, one being the tutorial, one being the level.

Timeline

Sprint 1

Design: Brainstorm the game idea, determine gameplay elements, and design mechanics.

Sprint 2 (MVI)

Implementation: Make a potion, refine an ingredient, and confirm a sale.

Aesthetics: Using temporary assets, but finalize the art style, create concept art, and design the scene layout.

Sprint 3

Design: Minor pivot in gameplay loop design (from multiple customers per day(level) to focusing on one per day(level)).

Implementation: Additional refinement options and a single potion with multiple steps, add customer interactions, and a dialogue system.

Aesthetics: Create assets for ingredients and tools, and start to replace the temporary assets in the scene.

Sprint 4

Implementation: Implement potion effects and their amplified versions, and start tracking potion outcomes.

Aesthetics: Design UI elements for potion brewing, customer requests, and dialogue windows; improve environment visuals.

Sprint 5

Implementation: Integrate multiple ingredients and refinement combinations, implement the tutorial customer flow, and finalize core game logic.

Aesthetics: Replace remaining placeholder assets, polish transitions, and add sound effects for mixing, refining, and selling.

Sprint 6

Implementation: Add the first (main) customer level with unique potion requests, as well as Rush Hour Mode, refine balancing, and finalize all mechanical interactions.

Aesthetics: Final background details; ensure cohesive art direction and consistency across scenes.

Final

Implementation: Conduct full integration testing, fix bugs, and ensure all systems (crafting, selling, dialogue, progression) function smoothly.

Aesthetics: Apply final polish—refine UI, adjust colors and effects, and prepare the build for presentation.

Team Roles

Producer - Samuel Denton

Artist - Brady Marr

Main Gameplay Programmer - Kevin Zheng

Assistant Programmer/Bugfixing - Galen Sagarin

Gameplay

Core Mechanic

The core game mechanic centers around potion making, which integrates several sub-mechanics such as ingredient picking and refining. Players craft potions by following recipes from a potion book, selecting from a set of available ingredients that are gradually unlocked as the game progresses. Each potion has three levels of effect, or tiers, with higher-tier recipes requiring more ingredients and more complex refinement steps. The quality of the final potion depends on how carefully the player follows the recipe and completes the refinements; any mistakes along the way can result in a dud (useless) potion.

Secondary Mechanics

Order System

Customer Requests are delivered through dialogue between the customer and the player. During the interaction, the customer explains the specific effects they need from the potion.

Order Fulfillment occurs when a player completes a customer's order by delivering a crafted potion. This is accomplished through a drag-and-drop interaction, where the player picks up

the crafted potion and places it on the designated area of the counter. Successfully delivering the potion completes the order, triggering updates to the customer's feedback.

Rating is the assessment of how well a product fulfills a customer's request. Under the hood, the system compares the final effects of the brewed potion to the customer's stated requirements.

Resource Management

Time Constraint requires the player to complete the order within a set time frame. If the player fails to finish the potion before time runs out, the customer will simply leave the shop, and the day ends.

Gameplay Loop

The core of the game revolves around a simple yet engaging loop of potion crafting and delivery. Players start by receiving orders and identifying the desired effects for each potion. They then refine ingredients through tactile mechanics such as grinding and plucking to prepare them for brewing. Once ready, ingredients are added to the cauldron, where they are combined and mixed to create the potion. After brewing, the finished potion is handed to the customer by placing it in the designated delivery area.

Asset Pipeline

Art Process

Our Artist was given a list of assets that were needed and some kind of timeline as to when they should be done. Due to illness toward the end of the project, not *everything* got finished but all of the assets were made in-house for the project aside from the sun for the timer, and the rope holding up the tickets (See Appendix for details).

Audio Process

Any and all music and sound effects were pulled from free use open repositories, as they were needed by our development team (See Appendix for details).

Technical Design

Customer

The parent class for all customers, including special cases like the Wizard. Each Customer instance holds references to its dialogue files, such as introduction and results dialogue, and tracks the desired potion type and quality. It also evaluates the submitted potion and controls the sprite's entrance and exit animations, serving as the base for specific customer behaviors and interactions in the game.

Ingredient

The abstract parent class for all ingredients. Each instance holds references to sprites for its different refined versions, the refinement tools that can be used on it, and its refinement log. The class provides helper methods for checking usable tools, tracking refinement history, updating the sprite to match the current state, and inheriting refinement logs for cases like Flower or Goat Hoof, where the instance may change when refined by tools such as shears.

Mouse Behavior

This script records all mouse behaviors, including clicks, releases, and dragging. The tracked data is used by other scripts to perform various checks and interactions.

Potion

The parent class for all potions. Each instance holds references to sprites for different potion tiers and tracks the current tier level. It provides methods, used by the Cauldron script (described below), to control the potion's quality (tier) and update the corresponding sprites accordingly.

Cauldron

This script keeps track of the list of potions it can make and holds references to the ingredients added to the cauldron. It works with a secondary script to handle the stirring mechanics. Once stirring is complete, the Cauldron analyzes each ingredient's type and refinement level, calculates the resulting potion type and tier, and then instantiates the corresponding potion instance.

Refinement

The abstract parent class for all refinement tools. It defines the core methods that each child tool must implement, including detecting when a refinement starts, performing actions during

refinement, handling post-refinement behavior, and resetting itself. Child classes of this script include **Grinder**, **Scissors**, **Hanger** & **Ropes**, and **Infuser**.

Ingredient Spawner

This script holds references to all insatiable ingredient prefabs and sets up references for newly instantiated instances. It is used by in-game spawners (such as ingredient sprites under the counter) and by certain refinement tools.

Customer Tracker

Inherited from the abstract **ChildTracker** class, which defines a list of tracked children, this script additionally manages ticket animations, remaining customer tracking, and order submission. It also handles customer behaviors during rush hour mode, including resetting customers and tickets and updating the timer.

Appendix

Resources

Graphics

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